The Business Value of Diversity Combined with Data Science

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Abstract

Strong diversity programs, sensitivity to diversity issues, and the disciplined use of data science can create measurable business value. Organizations often discover that the first visible symptom of a business problem appears through a diversity lens, while the durable solution comes from rigorous analysis.

This paper presents practical examples showing how diversity awareness, paired with SAS-based analytics, can improve retention, sharpen decision-making, reduce compliance risk, strengthen client relationships, and produce better science. Many organizations are aware of both the importance of diversity and the fact that problems sometimes manifest themselves as diversity issues, even when they have other causes.

It also explores areas where certain regulatory requirements may change and where other areas have become newly important.

For programmers, statisticians, business leaders, and decision-makers, the central point is straightforward: when collected well, governed well, and analyzed well, diversity becomes a business asset.

Introduction

Many business challenges begin with a concern that appears to involve diversity. In some cases, the concern turns out to be a symptom rather than the root cause. In other cases, diversity itself can be a source of strategic advantage because a broader range of experiences, perspectives, and demographic realities improves analysis, product design, client credibility, and organizational adaptability.

The examples in this paper illustrate both patterns. Some show how a diversity-related question led to the discovery of a different operational issue. Others show that diversity itself improved outcomes by

making research more representative, helping companies understand markets more accurately, or strengthening the organization's ability to attract and retain talent.

The role of data science is crucial. Well-structured data, longitudinal analysis, text mining, segmentation, forecasting, and statistical controls are what separate assumptions from evidence. For that reason, diversity becomes most valuable when it is connected to a disciplined analytic framework.

Childcare

An organization noticed that women in the organization were, on average, being paid less than men. Analysis using SAS broke salaries down by age, education, and experience, and the apparent gender pay discrepancy disappeared within comparable groups. Additional analysis then showed that many more women than men were leaving in their late twenties or early thirties, which reduced average female tenure and experience levels in the organization's workforce.

Using SAS Text Mining software on exit interviews revealed that childbirth and childcare issues were common reasons for departure. Voluntary turnover is costly because it leads to additional recruitment, training, and onboarding expenses, and it can also reduce continuity and operational effectiveness. The organization then established childcare centers at selected locations.

A later longitudinal analysis compared changes in voluntary turnover at locations with childcare centers against changes at comparable locations in the same geographic areas without such centers. There are many things happening in the economy not related to childcare that influence voluntary turnover: in a growing economy more jobs are available and people are more willing to take risks while in a declining economy people want to hang on to what they've got.

Therefore, the organization compared the before-and-after change in voluntary turnover in the locations where the childcare centers had been set up to the before-and-after change in voluntary turnover in other locations in the same geographic area, and they found a statistically significant decline in voluntary turnover in the locations that had childcare centers. Interestingly, many male employees also used the childcare centers.

This case remains powerful because it shows how information that surfaced during a diversity analysis can reveal a hidden business issue. The lesson for a SAS audience is equally important: careful segmentation, text analytics, and controlled before-and-after comparisons can convert a broad HR concern into a targeted, high-value business intervention.

Racial Diversity

In the early 2000s, a pharmaceutical organization had the perception that they were losing African-American sales representatives and that this might be due to a hostile work environment. Investigation showed that the company was not losing a disproportionate number of African-American sales representatives. Instead, it was losing an increasing number of younger sales representatives across racial groups.

Text mining of exit interviews using SAS software showed that many departing sales representatives had accepted higher-paying jobs at competing pharmaceutical companies. The company participated in a salary survey with other pharmaceutical companies. A statistical analysis of the salary survey showed that the company's compensation had fallen behind the market. The problem was not a hostile work environment; it was noncompetitive pay.

Again, attention to diversity created value by helping the organization ask the right question. Without the initial concern, management might not have recognized the compensation problem quickly enough. The broader analytic lesson is that diversity monitoring should not stop after a pattern is detected. It should continue until the organization understands the mechanism driving that pattern.

This remains relevant in the current environment because workforce analytics now sits closer to investor scrutiny, talent competition, and social disclosure expectations than it did a decade ago. McKinsey's research continues to reinforce the connection between diversity and stronger business outcomes, while newer reporting frameworks increasingly push organizations to support diversity narratives with evidence rather than assertions.

The context in this case was interesting. During the Clinton administration in the 1990's the administration proposed a health care program that would have led to large, centralized buying groups. As a result, pharmaceutical companies had cut their sales forces in anticipation of having to deal with a small number of central groups instead of a large number of individual doctors. When the proposed program was not enacted, the pharmaceutical companies needed to restaff their sales forces and thus were luring the younger sales reps (who cost less) from other pharmaceutical companies by offering higher salaries.

Pharmaceutical Research

An especially important example for a SAS audience comes from clinical research. A PharmaSUG 2024 paper by Donna Sattler, dsattler0716@gmail.com, *Advancing Diversity in Clinical Research: BMS Case Study on Self-Identified Sexual Orientation, Gender Identity (SOGI)*, highlighted that including a diverse set of subjects leads to more accurate and complete research because it reduces knowledge gaps. Diverse clinical populations help researchers detect differences that may affect smaller or previously underrepresented subgroups.

This point has become even more important. When patients feel seen, respected, and appropriately represented, they are more likely to remain engaged in research and care, which improves data quality and scientific validity. Better representation is not merely a social objective; it is good science, and increasingly, it is also a compliance and execution issue.

This is an area where SAS tools are highly relevant. Trial teams can use SAS to monitor enrollment against target demographics, detect underrepresentation early, analyze protocol deviations and dropout risk by subgroup, and support the preparation of auditable reporting packages. For statisticians, the practical issue is not only whether subgroup analysis is possible after the fact, but whether the study and data pipeline are designed early enough to make those analyses credible. As Sattler points out, "Diversity truly is just good science".

Sexual Orientation

A consulting organization had an active and successful LGBT (Lesbian Gay Bisexual Transgender) group. At one client, a client's LGBT group was struggling to gain traction and asked for help. Although the support was outside the formal contract, the consulting team connected the client group with successful internal LGBT groups in nearby areas, shared structural advice, and introduced the client team to other relevant organizations.

The client was very appreciative, and the effort helped strengthen the business relationship. This example shows that strong internal diversity efforts can become part of the value a company offers externally. In many professional settings, clients want not only technical competence but also practical insight into culture, employee engagement, and the governance of inclusion.

Retail Campaigns

While working with a hair care company, there was an expectation that an aging population would sharply increase sales of products designed to conceal gray hair. However, many women in the organization recognized that as the number of older women increased in society, attractive gray hair itself could become fashionable, and silver tones and related looks could become more desirable.

That insight helped the strategy and IT teams recalibrate demand forecasting more effectively. The example illustrates how a diverse workforce can improve decision-making by bringing lived experience into forecasting, product strategy, and interpretation of consumer trends.

For SAS professionals, this kind of case is a reminder that forecasting models improve when the business assumptions behind the model are challenged intelligently. Diverse teams do not replace analytics; they improve the quality of the questions that shape model design, feature selection, and interpretation.

Regulatory Matters

Private organizations and local government organizations with more than 100 employees, and federal government contractors with more than 50 employees are required to submit EEO-1 reports which show gender, race, and ethnicity in ten different job group categories. For many organizations, these reports are not just filing exercises; they are indicators of whether workforce patterns may draw regulatory attention or suggest the need for corrective analysis. A copy of the EEO-1 report is below.

NOTE: On May 14, 2026, the U.S. Equal Employment Opportunity Commission (EEOC) submitted a formal proposed rule to the White House Office of Management and Budget (OMB) and the Office of Information and Regulatory Affairs (OIRA) that will eliminate the requirement for the EEO-1 report. At this

point (October 1, 2026), however, the EEO-1 report is still required, so organizations would be well-advised to keep their data readily available. Also, state-level requirements for similar reports or data will still be in force. The prudent course is to keep workforce demographic data structured, validated, and audit-ready.

5. EEO-1 Form (To be submitted by the prime and any sub with 20% or more of the contract).

NAME OF FIRM:

5. Section D- EMPLOYMENT DATA

Employment at this establishment- Report all permanent full and part-time employees including apprentices and on-the-job trainees unless specifically excluded as set forth in the instructions. Enter the appropriate figures on all lines and in all columns. Blank spaces will be considered as zeros.

Job Categories		Number of Employees (Report employees in only one category)														
		Race/Ethnicity														
		Hispanic or Latino		Not-Hispanic or Latino												TOTAL COL. A-N
				Male						Female						
		Male	Female	White	Black or African American	Native Hawaiian or Other Pacific Islander	Asian	American Indian or Alaska Native	Two or more races	White	Black or African American	Native Hawaiian or Other Pacific Islander	Asian	American Indian or Alaska Native	Two or more races	
		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Executive/Senior Level officials and Managers	1.1															
First/Mid-Level Officials and Managers	1.2															
Professionals	2															
Technicians	3															
Sales Workers	4															
Administrative Support Workers	5															
Craft Workers	6															
Operatives	7															
Laborers and Helpers	8															
Service Workers	9															
TOTAL	10															
PREVIOUS YEAR TOTAL	11															

1. Date(s) of payroll period used (Omit on the Consolidated Report.)

Section E-ESTABLISHMENT INFORMATION (Omit on the Consolidated Report) What is the major activity of this establishment? (Be specific, i.e., manufacturing steel casings, retail grocer, wholesale plumbing supplies, title insurance, etc. Include the specific type of product or type of service provided, as well as the principal business or industrial activity.)

Section F- REMARKS-Use this item to give any identification data appearing on the last EEO-1 report which differs from that given above, explain major changes in composition of reporting units and other pertinent information

Section G- CERTIFICATION

Check One			
	1	☐	All reports are accurate and were prepared in accordance with the instructions. (Check on Consolidated Report only.)
	2	☐	This report is accurate and was prepared in accordance with the instructions.

Name of Certifying Official	Title	Signature Signature	Date
Name of person to contact regarding this report	Title	Address (Number and Street)	

Employee Morale and Productivity

A diverse workforce can improve decision-making, but it can also improve morale and productivity when employees feel comfortable bringing their whole selves to work. More motivated employees tend to contribute more consistently and collaborate more effectively, and organizations benefit from the wider range of perspectives available to solve problems and identify opportunities.

External rankings and employer reputation also matter. There are rankings that are published by other organizations. For example, Forbes publishes lists such as "America's Best Employers" and "Best Employers for Diversity," which are based on employee surveys and other factors like company perks and benefits. Having high rankings can improve retention and give the organization a better opportunity to hire the best employees.

This is another area where data science can add value beyond basic reporting. Internal surveys, workforce movement data, performance indicators, and retention patterns can be combined to detect emerging issues, identify successful teams or practices, and estimate the likely effect of interventions before problems become expensive.

A solid diversity tracking system using SAS, combined with a comprehensive HR database, can help an organization monitor its diversity on several different axes, produce the statistics to submit to the companies doing the rankings, and produce any other statistics that the organization would like to track. Likewise, a sophisticated diversity tracking system can pinpoint areas where improvements or corrective actions might be necessary. Similarly, internal employee attitude surveys can combine the trends in the organization's workforce with employee experiences to allow the organization to use advanced data science techniques and detect upcoming issues and find areas where positive results are taking place.

New Department of Transportation Family-Friendly Programs for Mass Transit Organizations

Secretary of Transportation Sean Duffy has made a series of funding announcements prioritizing "family-friendly" mass transit investments. The United States Department of Transportation (USDOT) is now prioritizing funding for local transit infrastructure and corridors that better support families and childcare networks.
The Federal Transit Administration (FTA) has announced funding programs in the following areas:

Transit-Oriented Development (TOD) Planning: Proposals that demonstrate how they will integrate childcare centers and recreational facilities directly into or near public transportation hubs. The goal is to design corridors that naturally connect working parents to jobs, schooling, and healthcare.

Station Accessibility and Modernization: Funding for older or heavily trafficked rail stations to physically modify or relocate infrastructure to accommodate families. A core metric used to judge applications is how well the layout accommodates parents traveling with young children, specifically factoring in ease of use for strollers.

The Family Friendly Transit Dashboard: The USDOT is launching a public tracking metric to score mass transit systems nationwide on how effectively they cater to American families. The core benchmarks being measured include station cleanliness, reliability, safety, and universal accessibility.

Increased Emphasis on intellectual diversity

There has been an increased focus on the importance of intellectual diversity, or "viewpoint diversity", in many areas, especially in college faculties and research-oriented environments. The push for intellectual diversity on college campuses often centers on the idea that a university should be a "marketplace of ideas" where students are exposed to a broad spectrum of political, social, and philosophical viewpoints. As illustrated in the examples above, organizations with diverse groups of employees have been able to use the insights gained from their diversity to weather challenges, improve results, and as mentioned earlier, enhance employee morale.

Having a diverse group of viewpoints can help organizations gain additional perspectives as well as opening up links to people in other areas who may be able to help the organizations. Outside funding could be dependent on having a diversity of ideas and providing exposure to opposing views can sharpen people's analytical skills. In many areas, like the example in pharmaceutical research mentioned earlier in this paper, "diversity is just good science" and this can apply to intellectual viewpoints as well as demographic traits.

Conclusion

Diversity is a business asset, and awareness of diversity can lead to improved business outcomes. Problems that might not otherwise have been apparent can be detected by using advanced data science techniques, and diverse organizations can use a wider range of perspectives when making decisions.

The examples in this paper show several ways this happens. Attention to diversity can uncover hidden operational problems such as childcare-related turnover or noncompetitive compensation. It can improve product strategy and forecasting. It can strengthen client relationships. It can improve morale, productivity, and talent outcomes. In life sciences, it can improve the quality and credibility of research itself. Organizations that build disciplined diversity analytics capabilities now will be better positioned to manage risk, improve decisions, and create value.

Author Biography

Stephen B. Sloan is a senior data science and analytics professional with experience across project management, data management, and statistical analysis in human resources, supply chain, finance, marketing, insurance, life sciences, and manufacturing for both private and government clients. He has worked extensively with SAS technologies and has collaborated with many talented professionals connected to SAS Institute.

Stephen has presented more than 100 times at 51 SAS conferences and has published in professional journals. He holds a B.A. cum laude with Honors in Mathematics from Brandeis University, M.S. degrees in Mathematics and Computer Science from Northern Illinois University, an MBA from the Stern School of Business at New York University, where he finished first in his class, and a graduate certificate in Financial Analytics from Stevens Institute.

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